

A Single Stepsize Suffices for Unprojected Linear TD(0)

Simultaneous Robust and Fast Rates via Polyak–Ruppert Averaging

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Setting: Linear TD(0) with Markovian sampling

A fixed policy induces a Markov reward process and a single **dependent** trajectory:

$$|\mathcal{S}| \leq \infty, \quad s_0, s_1, s_2, \dots, \quad r_t = r(s_t, s_{t+1}) \leq r_\infty, \quad Z_t = (s_t, s_{t+1}).$$

Policy evaluation under linear function approximation asks for

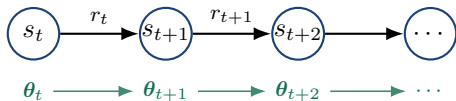
$$V^\mu(s) = \mathbb{E} \left[\sum_{k=0}^{\infty} \gamma^k r_k \middle| s_0 = s \right], \quad V_\theta(s) = \phi(s)^\top \theta \quad (\phi(s) \in \mathbb{R}^d, \|\phi(s)\| \leq \phi_\infty).$$

Unprojected linear TD(0) uses the TD direction

$$\theta_{t+1} = \theta_t + \eta_t \mathbf{g}(\theta_t, Z_t), \quad \mathbf{g}(\theta, Z_t) = (r_t + \gamma \phi(s_{t+1})^\top \theta - \phi(s_t)^\top \theta) \phi(s_t).$$

The stationary TD direction and TD fixed point are

$$\bar{\mathbf{g}}(\theta) := \mathbb{E}_{Z \sim \pi_Z} [\mathbf{g}(\theta, Z)], \quad \theta^* : \quad \bar{\mathbf{g}}(\theta^*) = \mathbf{0}.$$



Performance measure: TD potential and two regimes

For $v \in \mathbb{R}^{|\mathcal{S}|}$, define the stationary value norm and Dirichlet seminorm

$$\|v\|_D^2 := \sum_s \pi(s) v(s)^2, \quad \|v\|_{\text{Dir}}^2 := \frac{1}{2} \sum_{s, s'} \pi(s) P^\mu(s, s') (v(s) - v(s'))^2.$$

We measure error by the TD potential

$$f(\boldsymbol{\theta}) - f(\boldsymbol{\theta}^*) = (1 - \gamma) \|\mathbf{V}_{\boldsymbol{\theta}} - \mathbf{V}_{\boldsymbol{\theta}^*}\|_D^2 + \gamma \|\mathbf{V}_{\boldsymbol{\theta}} - \mathbf{V}_{\boldsymbol{\theta}^*}\|_{\text{Dir}}^2.$$

It has curvature around the TD fixed point $\boldsymbol{\theta}^*$:

$$f(\boldsymbol{\theta}) - f(\boldsymbol{\theta}^*) \geq \omega \|\boldsymbol{\theta} - \boldsymbol{\theta}^*\|^2, \quad \omega := (1 - \gamma) \lambda_{\min}(\boldsymbol{\Phi}^\top \mathbf{D} \boldsymbol{\Phi}) > 0.$$

Two finite-time regimes

Robust rate

curvature-free

$$\tilde{\mathcal{O}}(1/\sqrt{T})$$

Fast rate

curvature-dependent

$$\tilde{\mathcal{O}}(1/(\omega T))$$

Why stability is hard under Markovian sampling

Two coupled difficulties.

- **Not i.i.d.:** $Z_t = (s_t, s_{t+1})$ is generated by the same trajectory that produced θ_t . Thus, in general,

$$\mathbb{E}[g(\theta_t, Z_t) \mid \mathcal{F}_{t-1}] \neq \bar{g}(\theta_t).$$

- **Self-amplification:**

$$\|g(\theta_t, Z_t)\| \leq r_\infty \phi_\infty + 2\phi_\infty^2 \|\theta_t\|.$$

A loose bound on $\|\theta_t\|$ makes the update bound loose, which feeds back into θ_{t+1} .

What prior work uses

- **Projection:** (Bhandari et al. 2018; Liu and Olshevsky 2021). Stable by design, but changes TD algorithm and requires prior knowledge of a suitable projection radius.
- **Curvature / contractive stability:** (Srikant and Ying 2019; Patil et al. 2023; Samsonov et al. 2024; Mitra 2025; Chandak and Borkar 2025; Durmus et al. 2025; Li et al. 2026). Projection-free or lightly modified, but the rate, burn-in, or tuning can depend on curvature.

Question: Can plain unprojected TD(0) achieve high-probability stability and simultaneous robust and fast rates without knowledge of ω ?

Main results: stability and simultaneous rates

Use a single decaying stepsize

$$\eta_t = \frac{1}{c \tau_{\text{mix}} \phi_\infty^2} \frac{1}{\sqrt{t+1} \log(t+3)}.$$

Let

$$R_{\text{base}} = \max \left\{ \|\theta_0 - \theta^*\|, \|\theta^*\|, \frac{r_\infty}{\phi_\infty} \right\}.$$

Define the weighted Polyak–Ruppert average

$$S_T = \sum_{t=1}^T \eta_{t-1}, \quad \bar{\theta}_T = \frac{1}{S_T} \sum_{t=1}^T \eta_{t-1} \theta_{t-1}.$$

Mixing assumption

Assume $Z_t = (s_t, s_{t+1})$ is irreducible and aperiodic with stationary law π_Z . Then,

$$\max_{z \in \mathcal{Z}} \|P_Z^k(z, \cdot) - \pi_Z\|_{\text{TV}} \leq 4 \cdot 2^{-k/\tau_{\text{mix}}}, \quad k \geq 0.$$

Main theorem

For a sufficiently large c depending on δ , set $R_{\text{max}} = \rho(c, \delta) R_{\text{base}}$. With probability at least $1 - \delta$, for all $T \geq 1$,

$$\sup_{t \geq 0} \|\theta_t\| \leq R_{\text{max}}, \quad f(\bar{\theta}_T) - f(\theta^*) \leq \tilde{\mathcal{O}} \left(R_{\text{max}}^2 \min \left\{ \frac{\tau_{\text{mix}}^2 \phi_\infty^4}{\omega T}, \frac{\tau_{\text{mix}} \phi_\infty^2}{\sqrt{T}} \right\} \right).$$

Stability proof I: make the loop self-bounding

Induct on the past event

$$\text{IH}_{t-1} := \left\{ \max_{0 \leq i < t} \|\boldsymbol{\theta}_i\| \leq R_{\max} \right\}.$$

Since $\|\boldsymbol{\theta}_t\| \leq \|\boldsymbol{\theta}_t - \boldsymbol{\theta}^*\| + \|\boldsymbol{\theta}^*\|$, we can use $\|\boldsymbol{\theta}_t - \boldsymbol{\theta}^*\|$ to control $\|\boldsymbol{\theta}_t\|$. Expanding $\|\boldsymbol{\theta}_t - \boldsymbol{\theta}^*\|^2$ gives

$$\|\boldsymbol{\theta}_t - \boldsymbol{\theta}^*\|^2 = \|\boldsymbol{\theta}_0 - \boldsymbol{\theta}^*\|^2 + \sum_{i=0}^{t-1} \eta_i^2 \|\mathbf{g}_i\|^2 + 2 \sum_{i=0}^{t-1} \eta_i \langle \bar{\mathbf{g}}(\boldsymbol{\theta}_i), \boldsymbol{\theta}_i - \boldsymbol{\theta}^* \rangle + 2 \sum_{i=0}^{t-1} \eta_i h_i(Z_i),$$

where

$$h_i(z) := \langle \mathbf{g}(\boldsymbol{\theta}_i, z) - \bar{\mathbf{g}}(\boldsymbol{\theta}_i), \boldsymbol{\theta}_i - \boldsymbol{\theta}^* \rangle.$$

The mean drift helps:

$$\langle \bar{\mathbf{g}}(\boldsymbol{\theta}_i), \boldsymbol{\theta}_i - \boldsymbol{\theta}^* \rangle = -[f(\boldsymbol{\theta}_i) - f(\boldsymbol{\theta}^*)] \leq 0.$$

Under IH_{t-1} , the quadratic term is controlled because $\sum_i \eta_i^2 < \infty$. The only hard term is the Markov bias

$$B_t := \sum_{i=0}^{t-1} \eta_i h_i(Z_i).$$

Stability proof II: Poissonize the Markov bias

Freeze θ_i . Since $\mathbb{E}_{Z \sim \pi_Z}[h_i(Z)] = 0$, solve the Poisson equation

$$u_i - P_Z u_i = h_i, \quad u_i(z) = \sum_{\ell=0}^{\infty} (P_Z^\ell h_i)(z), \quad \|u_i\|_\infty \leq 16\tau_{\text{mix}} \|h_i\|_\infty.$$

Then each Markov-noise term decomposes as

$$h_i(Z_i) = \underbrace{u_i(Z_{i+1}) - (P_Z u_i)(Z_i)}_{\text{martingale difference}} + \underbrace{u_i(Z_i) - u_i(Z_{i+1})}_{\text{telescoping remainder}}.$$

Under IH_{t-1} ,

$$\|h_i\|_\infty \lesssim \phi_\infty^2 R_{\text{max}}^2, \quad \|u_i\|_\infty \lesssim \tau_{\text{mix}} \phi_\infty^2 R_{\text{max}}^2.$$

Two key implications

- The first term is controlled by Pinelis' inequality from bounded differences.
- The second term is rewritten by Abel summation; the moving u_i 's are controlled by Lipschitz continuity and $\theta_i - \theta_{i-1} = \eta_{i-1} \mathbf{g}_{i-1}$.

Stability proof III: close the bootstrap

Controlling the martingale term and the telescoping remainder gives

$$B_t \lesssim \left(\frac{\rho^2}{c} + \frac{\rho^2}{c^2} \right) R_{\text{base}}^2.$$

Combining this with the quadratic gradient term yields

$$\|\theta_t\|^2 \leq \left(4 + \frac{A_1(\delta)\rho^2}{c} + \frac{A_2\rho^2}{c^2} \right) R_{\text{base}}^2,$$

where $A_1(\delta)$ and A_2 are problem-independent constants.

Choose

$$c > \frac{A_1(\delta) + \sqrt{A_1(\delta)^2 + 4A_2}}{2}, \quad \rho = \frac{2c}{\sqrt{c^2 - A_1(\delta)c - A_2}}.$$

Then

$$4 + \frac{A_1(\delta)\rho^2}{c} + \frac{A_2\rho^2}{c^2} \leq \rho^2.$$

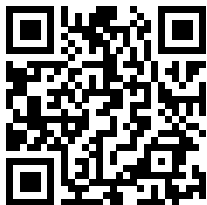
Bootstrap conclusion

$$\|\theta_t\|^2 \leq \rho^2 R_{\text{base}}^2, \quad \text{IH}_{t-1} \implies \text{IH}_t, \quad \sup_{t \geq 0} \|\theta_t\| \leq \rho R_{\text{base}}.$$

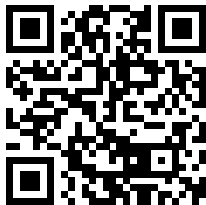
Thank you

Thank you! Any questions?

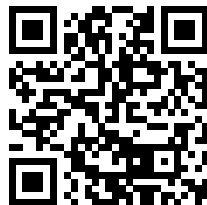
I would be happy to take any questions.



Slides



Paper



Poster

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